

Technical Q&A for NS1 NEON SMBA PRD NS1-SMBA-REQ-0001 Rev D

Question: What is SMBA's approach to radiometric sensitivity?

Answer:

Radiometric sensitivity (noise equivalent delta temperature (NEDT)) is defined as the standard deviation of the radiometer output in kelvins (K) when the antenna is viewing a 300 K uniform and stable target. The NEDT is computed as the root sum square of the thermal noise and 1/f noise contributions per the equation below.

$$NEDT_{total} = [NEDT_{thermal}^2 + NEDT_{1/f}^2]^{1/2}$$

Thermal noise, also known as white noise, is noise with a uniform power spectral density.

1/f noise, also known as pink noise or flicker noise, is noise with a power spectral density with a negative slope up to -1.

The 1/f noise ratio is defined as the power ratio of the 1/f noise to total noise, as computed in accordance with D0001-M01-S01-001 NOAA ATMS SDR ATBD Section 4.

NEDT computations apply to the instantaneous field of view (IFOV) and is computed in accordance with D0001-M01-S01-001 NOAA ATMS SDR ATBD Section 4 (<https://www.star.nesdis.noaa.gov/jpss/ATMS.php>).

The SMBA PRD has separate requirements for the thermal noise contribution to NEDT (to allow for scaling with integration time and bandwidth) and the 1/f noise contribution (specified as a ratio of 1/f noise to the total noise). Total NEDT will be reported in CDRL documentation.

Question: Why was heritage channel $f_c = 51.76$ GHz (ATMS Ch 4) removed and lowest F-band channel modified from $f_c = 114.5$ GHz to $f_c = 114.75$ GHz from the heritage set of channels?

Answer: These channels are in unprotected spectrum. The heritage channel list was changed so that the SMBA PRD does not require channels in unprotected spectrum.

Per the International Telecommunication Union (ITU) Radio Regulations 2024 (<https://www.itu.int/pub/R-REG-RR-2024>) page RR5-151, the frequency range of 51.4 – 52.4 GHz is not allocated to “Earth Exploration-Satellite (passive)”, and therefore is not considered protected spectrum for the SMBA instrument. Therefore, heritage channel $f_c = 51.76$ GHz was removed.

Per the International Telecommunication Union (ITU) Radio Regulations 2024 (<https://www.itu.int/pub/R-REG-RR-2024>) page RR5-159, the frequency range of 111.8 – 114.25 GHz is not allocated to “Earth Exploration-Satellite (passive)”, and therefore is not considered protected spectrum for the SMBA instrument. Priority was given to preserve channel bandwidth, therefore the center frequency was revised to avoid the unprotected spectrum.

Question: What is SMBA’s approach to calibration accuracy?

Answer: The SMBA PRD approach to calibration accuracy has been revised to be in accordance with the definitions of uncertainty from the National Institute of Standards and Technology (NIST), Bureau International des Poids et Mesures (BIPM), Joint Polar Satellite System (JPSS), NASA science and European missions. The revisions also bring the microwave definitions into alignment with the visible and short-, medium-, and long-wave infrared (IR) definitions. See below references:

1. Joint Committee for Guides in Metrology. (2012). *International vocabulary of metrology: Basic and general concepts and associated terms (VIM)* (3rd ed., JCGM 200:2012). Bureau International des Poids et Mesures. <https://www.bipm.org/en/committees/jc/jcgm/publications>
2. NIST/SEMATECH e-Handbook of Statistical Methods, <http://www.itl.nist.gov/div898/handbook/>
3. E. Kim et al., "An Evaluation of NOAA-20 ATMS Instrument Pre-Launch and On-Orbit Performance Characterization," in *IEEE Transactions on Geoscience and Remote Sensing*, vol. 60, pp. 1-13, 2022, Art no. 5302813, doi: 10.1109/TGRS.2022.3148663.

Uncertainty is defined as the total uncertainty comprised of the accuracy contributions and the precision contributions, as conceptually shown in Equation 1.

Accuracy is defined as the systematic portions of uncertainty that are constant or vary in a predictable manner. Also known as bias. Bias is the statistical mean of the uncertainty and can be signed positive or negative.

Precision is defined as random portions of uncertainty that vary in an unpredictable manner. Random error is one-sigma standard deviation of the uncertainty.

Calibration uncertainty of SMBA is defined as the difference between the brightness temperature inferred from the microwave radiometer (referred to the antenna collecting aperture) and the true brightness temperature.

In a typical total power radiometer using a two-point calibration scheme, the system calibration uncertainty, E_{unc} , is conceptually described by Equation 1 below—i.e., it includes contributions from accuracy uncertainty and precision uncertainty. Different radiometer architectures and calibration schemes may necessitate different computations.

Equation 1: $E_{unc} = f(E_{acc}, E_{prec})$

Equation 2: $E_{acc} = \delta E_{hot} + (1 - \delta)E_{cold} + E_{sys}$

Equation 3: $E_{prec} = \sqrt{E_{prec,hot}^2 + E_{prec,cold}^2 + E_{prec,sys}^2}$

Equation 4: $\delta = (TB_{scene} - TB_{cold}) / (TB_{hot} - TB_{cold})$

E_{acc} : total calibration accuracy

E_{hot} : calibration accuracy related to hot reference

E_{cold} : calibration accuracy related to cold reference

E_{sys} : calibration accuracy in radiometer calibration transfer function

E_{prec} : total calibration precision; the 1 sigma root sum square (RSS) of the contributing individual precisions

TB_{hot} : measured brightness temperature of the hot reference

TB_{cold} : measured brightness temperature of the cold reference

TB_{scene} : measured brightness temperature of the observed scene

Total calibration precision, E_{prec} , excludes precision uncertainty from NEDT. Therefore, requirements in Section 4.3 System Calibration Uncertainty exclude uncertainty from NEDT.

Question: How is polarization defined?

Answer:

SMBA quasi-horizontal and -vertical polarizations are defined by the following two equations:

Quasi Horizontal Polarization (QH) = $TV \sin^2 \theta + TH \cos^2 \theta$

Quasi Vertical Polarization (QV) = $TV \cos^2 \theta + TH \sin^2 \theta$

Note that from the equations above, the QV and QH reduce to TV (with electric field vector perpendicular to the orbital velocity direction) and TH (with electric field parallel to the orbital velocity direction), respectively, at nadir when theta is zero degree. Here theta is the scan-angle (from nadir). TV and TH are the brightness temperatures of the vertically polarized and horizontally polarized components, respectively, following the conventional definition. The vertical polarization is the component whose electric field is lying in the plane of incidence, and the horizontal polarization is the component whose electrical field is perpendicular to the plane of incidence. In other words, the SMBA instrument receives a linear combination of the pure vertical and horizontal components. The total electric field vector of the received wave (either QV, or QH) is lying in a plane

perpendicular to the propagation direction and making a polarization angle Φ_p (in degrees), with respect to a reference (zero degree) line L, which is the intersection of the tangent (at beam center) plane and the plane of incidence. The total electric field vector rotates (i.e., the polarization angle changes) with the nadir angle Θ equal to 0 degrees. For Quasi Vertical, $\Phi_p = \Theta$. For Quasi Horizontal Polarization, $\Phi_p = 90 - \Theta$.

Polarization twist angle is defined as the mechanical angular misalignment between the feed horn polarization orientation and optical polarization reference.

Question: Can the government clarify the definition of nonlinearity?

Answer: Nonlinearity is the departure from the expected response of an ideal linear radiometer. Channel nonlinearity is determined from a second-order or higher fit to the calibration accuracy data obtained during thermal-vacuum calibration testing. Residual nonlinearity is the residual error after applying a nonlinearity correction to the individual calibration accuracy points.

Question: When the spacecraft is performing orbit correction, is the instrument expected to operate through the maneuver?

Answer: Yes, the instrument is required to remain in Operational Mode (does not park the scan mirror) without damage to the instrument during spacecraft orbit corrections.

Question: Will be the SMBA instrument be powered ON or OFF during launch?

Answer: PRD is written to be inclusive of instrument designs that require being powered ON and instrument designs that require being OFF at launch.

Question: Where are the instrument temperature set points for worst-case hot operational and cold operational, survival defined?

Answer: PRD does not require test range temperatures for the instrument. Test range temperatures are defined by the instrument provider given the instrument design, the space-environment, spacecraft accommodations, etc. The instrument provider will communicate through the instrument ICD to the spacecraft provider the instrument temperature needs.

Question: Are there any new TBXs that affect the instrument design, interface with the spacecraft, or testing to be aware of?

Answer: No new TBXs that affect instrument design, interface with spacecraft, or testing in PRD Rev D. Some TBX were resolved (e.g., SRFs). Some TBXs were eliminated when the requirement was deleted. PRD Rev D TBX table has a due date column with a closure event.